Predicting Consumer Complaint Disputes in Finance using Machine Learning (AIOPS)

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***Abstract*— Consumer complaints in the financial sector present challenges including reputations damage and financial losses. This paper proposes a novel approach for predicting disputed complaints using a scalable machine-learning pipeline. We leverage historical complaint data, employing feature engineering techniques to extract insights from structured and unstructured complaint text. Our model, built using PySpark ML, achieves high accuracy in classifying newly registered complaints as likely to be disputed or not. Automated retraining using Airflow ensures continued effectiveness. Real-world dataset evaluation demonstrates the performance of traditional methods, enabling proactive intervention and mitigating risks. This study underscores the value of machine learning in enhancing complaint management processes in finance, with potential applicability across industries.**

Keywords—Machine learning, deep learning, NLP, Classifier, Artificial Intelligence, Big Data, Consumer Complaints, Predictive Modeling, Machine Learning, Financial Sector, Scalable Product

I. INTRODUCTION

The financial sector operates within a complex landscape of regulations and consumer expectations. Effectively managing consumer complaints is fundamental to upholding trust, fostering positive customer experiences, and ensuring compliance with regulatory requirements. However, traditional approaches to analyzing consumer complaints often fail to capture the nuanced nature of dissatisfaction and potential harm, leading to missed opportunities for intervention and resolution. In response to these challenges, this paper presents a novel approach to predicting disputed consumer complaints in the financial sector using machine learning (ML) techniques.

The management of consumer complaints [8] in the financial sector has historically relied on traditional methods, regulatory frame works [1], and data-driven analyses. However, these approaches have limitations in accurately identifying and prioritizing complaints, with a high likelihood of escalating into disputes. Moreover, the sheer volume of complaints received by financial institutions presents a significant hurdle in manually identifying critical issues for timely resolution.

Previous literature has highlighted challenges in effectively analyzing consumer complaints, including reliance on simplistic metrics, difficulty in identifying potential disputes, and inadequate methods for prioritizing resources. Additionally, traditional approaches often lack the capability to extract insights from unstructured complaint text, limiting the understanding of customer sentiment and the root causes of dissatisfaction.

The motivation for this work stems from the need to address these challenges and improve the effectiveness of consumer complaint management in the financial sector [19]. The primary objectives of this paper are to develop a scalable ML pipeline for predicting disputed consumer complaints, enhance proactive dispute management [13], and provide deeper insights into customer sentiment and regulatory compliance.

The contributions of this research include the development and implementation of sophisticated ML models for analyzing historical complaint data, as well as addressing the limitations of traditional complaint analysis methods. By leveraging ML techniques, financial institutions can proactively manage disputes, build stronger customer relationships, and navigate the regulatory landscape with greater confidence.

II. RELATED WORK

The current landscape of managing consumer complaints in the financial sector involves a combination of traditional methods, regulatory frameworks, and data-driven analyses. This section provides an overview of the existing systems and methodologies employed by financial institutions to handple consumer complaints and predict disputes.

A. MACHINE LEARNING MODELS

A. Model 1: Utilizing BoW with Naive Bayes: The initial baseline model, referred to as BoW-MNB, employs a multinominal Naive Bayes classifier [12] [2]. This model utilizes the reprocessing techniques and representations

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outlined previously, employing BoW and TF-IDF's document vectorization's methods.

```
# Model TfidfVectorizer
tokenizer = '(?u)\\b\\w\\w+\\b'
ngram = (1, 3)
max_features = 20000
vectorizer = TfidfVectorizer(
        token_pattern=tokenizer,
        ngram_range=ngram,
        max_features=max_features)

# Model: MNB
tfidf_mnb_model = make_pipeline(
        vectorizer, MultinomialNB())
```

Fig.1 Bag-of-Words with Naive Bayes

B.Model 2: Implementing BoW using a Multi-Layer Perceptron:

The second baseline model, denoted as BoWMLP, is a multi-layer perceptron (MLP) utilizing a similar (BoW) approach and TF-IDF document vectorization. The vocabulary size is set to |V | = 20, 000. This perceptron is implemented using the Tensorflow-Keras API. [6].

C. Bidirectional Long Short-Term Memory (Bi-LSTM) Model:

The subsequent model is a Word-based Long Short-Term Memory (LSTM) network, featuring a trainable input-word embeddings layer with an embedding dimension of 100 (E100- Bi-LSTM).This model has a similar vocabulary size |V | = 20, 000, and also utilizes the TensorFlow/Keras deep learning APIs.

```
# BoW-MLP model
output_classes = 11 # dataset classes
dense_nodes = 128   # hyperparam
dropout = 0.5       # hyperparam

model = Sequential()
model.add(Dense(dense_nodes,
       activation='relu', input_shape=(20000,)))
model.add(Dropout(dropout))
model.add(Dense(dense_nodes, activation='relu'))
model.add(Dense(
       output_classes, activation='softmax'))
```

Fig. 2 Bag-of-Words with ML

It achieves a**83.1%** classification accuracy on the test data. Fig. 3 illustrates the training progress of the Bi-LSTM model over 10 epochs.

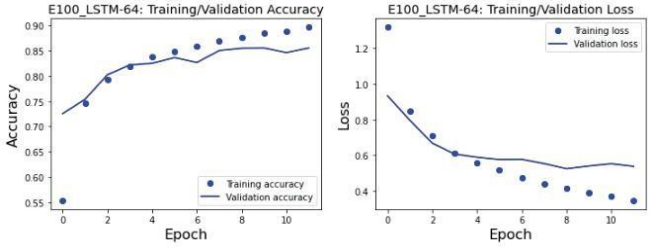


Fig .3 (Bi-LSTM) model losses and accuracy

In Fig. 4, the classification report depicts the precision, recall, and F-1 accuracy measures of the model for each of the output classification categories.

	precision	recall	f1-score	support
Bank account or service	0.91	0.81	0.85	193
Consumer Loan	0.90	0.81	0.85	173
Credit card	0.86	0.89	0.87	285
Credit reporting	0.88	0.96	0.92	363
Debt collection	0.87	0.79	0.83	400
Money transfers	0.70	0.54	0.61	13
Mortgage	0.92	0.97	0.94	474
Other financial service	0.00	0.00	0.00	5
Payday loan	0.69	0.44	0.54	25
Prepaid card	0.61	0.94	0.74	18
Student loan	0.83	0.96	0.89	51
accuracy			0.88	2000
macro avg	0.74	0.74	0.73	2000
weighted avg	0.88	0.88	0.88	2000

Fig .4 Classification Report

III. Proposed Method

This section delineates the methodology and procedural steps undertaken in the research to achieve the defined objectives. The research involves the development and implementation of a machine learning solution for predicting consumer complaint disputes in the financial sector.

We acquired our study data from the Consumer Financial Protection Bureau (CFPB) website (https://www.consumerfinance.gov) [16] [5] [4]. The CFPB platform allows for filtering the dataset based on the time period. Consequently, we filtered and downloaded five datasets covering time spans of 30 days, 90 days, half a year, a full year, and three years on January 10, 2021. Table I displays the date ranges and the number of complaints for each filtered dataset. Each dataset comprises nine products. We opted to download data spanning from one year to three years ago because, three years prior, the dataset contained more than nine types of products. Logistic Regression and Support Vector Machine (SVM) classification models were formulated and implemented using Python code within a Jupyter Notebook environment. [14] [11]. Feature transformation and training-testing data were also performed [9].

TABLE I. **Filtered Datasets**

Dataset	Date Range	Number of Complain
1 Month	1 December 2021 to 31 December 2021	45,612
3 Months	1 October 2021 to 31 December 2021	130,956
6 Months	1 July 2021 to 31 December 2021	250,808
1 Year	1 January 2021 to 31 December 2021	440,591
3 Years	1 January 2019 to 31 December 2021	974,757

A. *Consumer Complaint Repository at CFPB: Unveiling Trends and Insights DATABASE*

[16]The collection of consumer complaints by the CFPB has been ongoing since late 2011. In 2024 alone, the bureau received approximately 4,955,881 complaints received in 2024 alone, contributing to a total of over 4,955,881, complaints amassed over the years (6 GB data) total. For this study, we accessed the CFPB dataset through the bureau's official website using an API [5] [10] [4], enabling us to obtain data up to the year 2024. Within this dataset, comprising 4,955,881 records, there are 4,954,087 instances containing non-empty complaint document texts.

B. *Data Set*

We chose the CSV format from the CFPB database,

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which provides data in both CSV and JSON formats. entry encompasses 18 fields, with focal points including ”complaint ID,” ”date received,” ”complaint description,” ”company,” ”state,” and ”ZIP code.” Additionally, essential fields such as”product,” ”sub-product,” ”issue,” and ”sub-issue” play pivotal roles in labeling. Noteworthy is the ”complaint description” field, housing consumer narratives, while the ”product” and ”issue” fields serve as crucial classification labels, as depicted in Fig. 5.

complaint_id	complaint_what_happened	product	issue
4971409	I have already filed a dispute on the incorrec...	Credit reporting, credit repair services, or o...	Problem with a credit reporting company's inve...
5037553	On XX/XX/21 at approximately XXXX am I was con...	Debt collection	Communication tactics
5020621	Hello, I have a loan with PNC bank, and it has...	Vehicle loan or lease	Managing the loan or lease
5014775	I had a car loan with XXXX XXXX. There was a C...	Credit reporting, credit repair services, or o...	Problem with a credit reporting company's inve...

Fig.5 Sample copy of CFPB database records

The consumer complaint narratives within the utilized subset comprise 57,892 documents, encompassing a total of 9,831,576 words and featuring a vocabulary of 36,724 unique words. The average document length stands at 204.3 words, with a minimum of 7 words and a maximum of 1,023 words in Fig. 6.

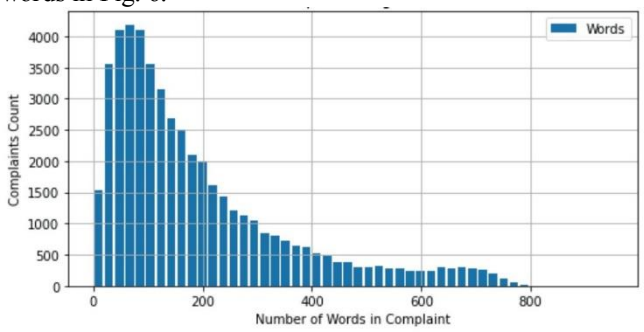


Fig .6 Complaint lengths, in words

It’s important to underscore that the consumer complaint [18] narratives housed in the CFPB database undergo inclusion solely upon the consumer’s consent. Precautionary Fig. 5. Sample copy of CFPB database records Fig. 6. Complaint length, in words measures are implemented to sanitize the data by removing personally identifiable information like names, addresses, and account numbers, thereby ensuring the preservation of privacy and confidentiality.

C. *Dataset’s size and distribution*

The data exhibits a skewed distribution across complaint categories. Among these, the category labeled ’Debt Collection’ stands out as the most prevalent, with a significant count of 18,445 complaints, constituting 27.27% of the total. In contrast, ’Money transfers’ and ’Other financial services’ make up approximately 2% and less than 1.5% of the total complaints, with 668 and 110 complaints respectively. Such distribution patterns are commonly encountered in real-world datasets. Figure 8 depicts a histogram illustrating the distribution of complaints across different product categories.

Fig.7 The histogram illustrates the distribution of complaints by product category.

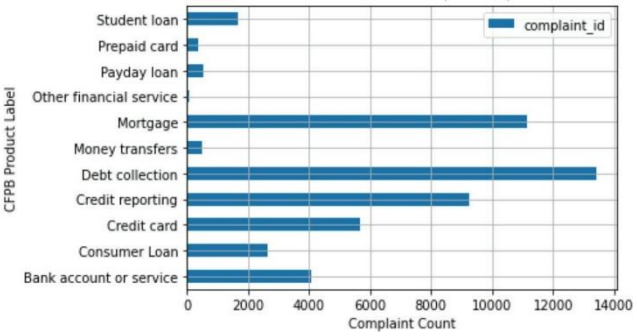


Fig .7 Product Label Histogram

D. *Model Pipeline*

A. *Data Ingestion Process*

1) Data Retrieval: The first step in our data ingestion pipeline was to retrieve relevant financial complaint data from external sources [17]. This involved accessing publicly available datasets or APIs provided by regulatory bodies or financial institutions.

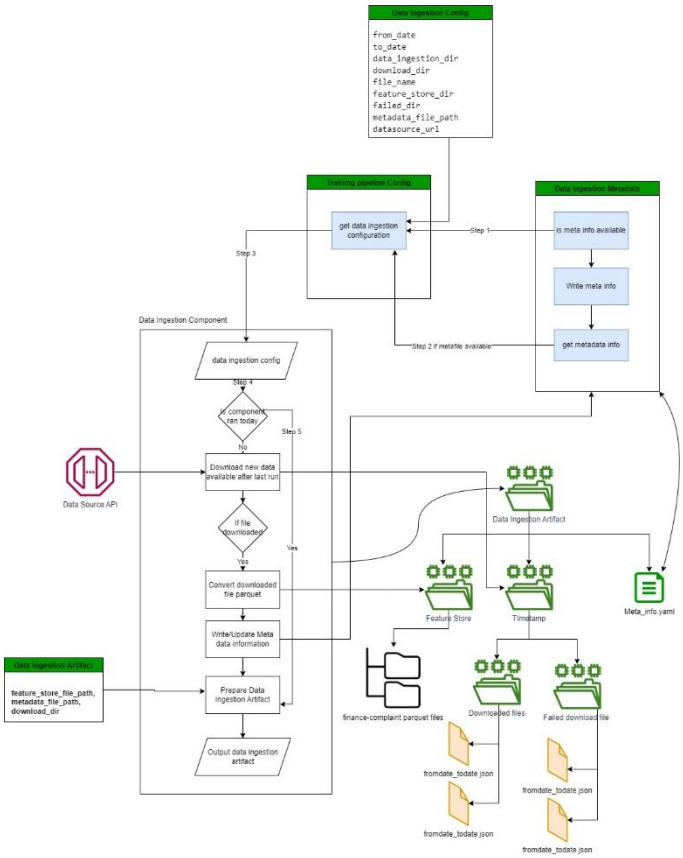


Fig .8 Data Ingestion

2) *Time Interval Determination*: Once the data sources were identified, we determined the time intervals for data retrieval based on the scope of our analysis. This step involved setting start and end dates to specify the time range of the data to be collected.

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3) *Download and Storage*: Using the specified time intervals, the data retrieval process was automated to download the relevant data files from the designated sources. These files were then stored in a local directory for further processing.

4) *Data Conversion*: The downloaded data, typically in JSON format, was converted to a more suitable format for analysis, such as Parquet. This conversion process facilitated faster data loading and improved efficiency during subsequent modeling steps.

5) *Metadata Management*: To keep track of the downloaded data and avoid redundant downloads, metadata about the downloaded files was maintained. This metadata included information such as the date range of the data, file paths, and download status.

6) *Error Handling and Retry Mechanism*: Robust error handling mechanisms were implemented to deal with potential issues during the data ingestion process. In cases of failed downloads or network issues, a retry mechanism was employed to ensure data completeness and integrity.

7) *Data Ingestion Automation:* Finally, the entire data ingestion process was automated to run periodically, ensuring that the dataset remained up-to-date with the latest available information.

B. Data Validation

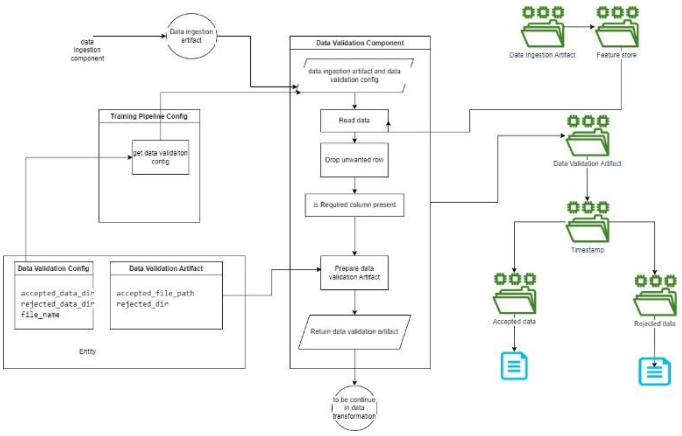


Fig .9 Data Validation

1) *Data Integrity Check*: The first step in data validation involves checking the integrity of the dataset obtained from the data ingestion process. We utilize a custom data validation framework to perform various checks, including:

a) *Missing Data Analysis*: We analyze the dataset to identify columns with missing values and compute the algorithm calculates the proportion of missing values in each column. Columns exhibiting a significant proportion of missing values are identified and marked for additional scrutiny.

b) *Unwanted Columns Detection*: We identify and remove unwanted columns from the dataset based on predefined criteria. These criteria include columns deemed irrelevant for the predictive modeling task or columns with a high percentage of missing values exceeding a specified threshold.

2) *Schema Validation*: Ensuring that the dataset conforms to the expected schema is essential for maintaining data consistency and facilitating downstream analysis. We validate the dataset against a predefined schema, which includes:

a) *Required Columns Check*: We verify the presence of all required columns specified in the schema. Any missing required columns are flagged as errors and require resolution before further processing.

b) *Unique Values Analysis*: We analyze each column to determine the number of unique values and the percentage of missing values. This analysis provides insights into the distribution and quality of the data, guiding subsequent preprocessing steps.

3) Data Preprocessing: Following data validation, the dataset undergoes preprocessing to prepare it for model training and evaluation. Preprocessing steps include:

a) *Unwanted Column Removal*: Unwanted columns and columns with a high percentage of missing values are dropped from the dataset to improve data quality and reduce noise.

b) **Data Imputation**: Missing values may be imputed using appropriate techniques such as mean imputation or interpolation to preserve valuable information and maintain dataset completeness.

4) Result Interpretation: The results of the data validation process provide insights into the quality and suitability of the dataset for predictive modeling. Any anomalies or issues identified during validation are documented and addressed, ensuring the reliability and robustness of the predictive models developed using the dataset.

C. Data Transformation

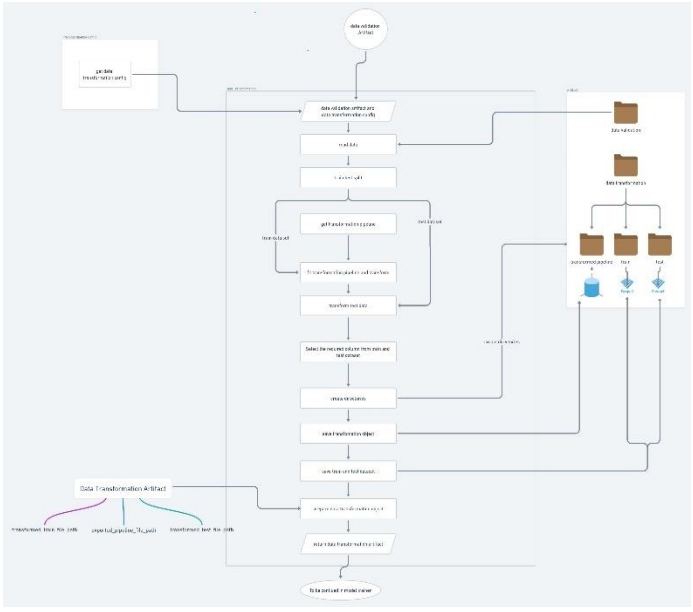


Fig .10 Data Transformation

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We introduce our custom data transformation approach designed to address these challenges effectively.

1) *Custom Transformation Components*:

1) *Frequency Encoder*:

a) *Description*: Frequency Encoder is a custom PySpark transformer designed to encode categorical variables based on their frequency in the dataset. It replaces categorical values with their respective frequencies, providing a more informative representation of the data.

b) *Implementation*: We demonstrate the implementation of Frequency Encoder using PySpark, highlighting its role in preprocessing categorical features.

2) Derived Feature Generator:

a) *Description*: Derived Feature Generator is a custom transformer that generates derived features from existing data attributes. It extracts additional information from raw features, enhancing the predictive power of machine learning models.

b) *Implementation*: We discuss how Derived Feature Generator is utilized to create new features based on domain knowledge and data insights.

3) Frequency Imputer:

a) *Description*: Frequency Imputer is a custom estimator employed to impute missing values in categorical variables using the most frequent category. It ensures robustness against missing data and preserves the integrity of the dataset.

b) *Implementation*: We illustrate the usage of Frequency Imputer to handle missing categorical values, emphasizing its role in data preprocessing.

2) *Integration into the Data Transformation Pipeline*:

We describe how the custom transformation components are seamlessly integrated into a comprehensive data transformation pipeline. This pipeline encompasses a series of preprocessing steps, including standard scaling, one-hot encoding, tokenization, hashing, and TF-IDF transformation, in addition to the custom components mentioned above.

3) *Experimental Evaluation:*

We present experimental results showcasing the effectiveness of our custom data transformation approach. Through comparative analysis and performance metrics, we demonstrate the impact of custom transformation components on data quality, feature representation, and the predictive performance of machine learning models.

D. Model Trainer

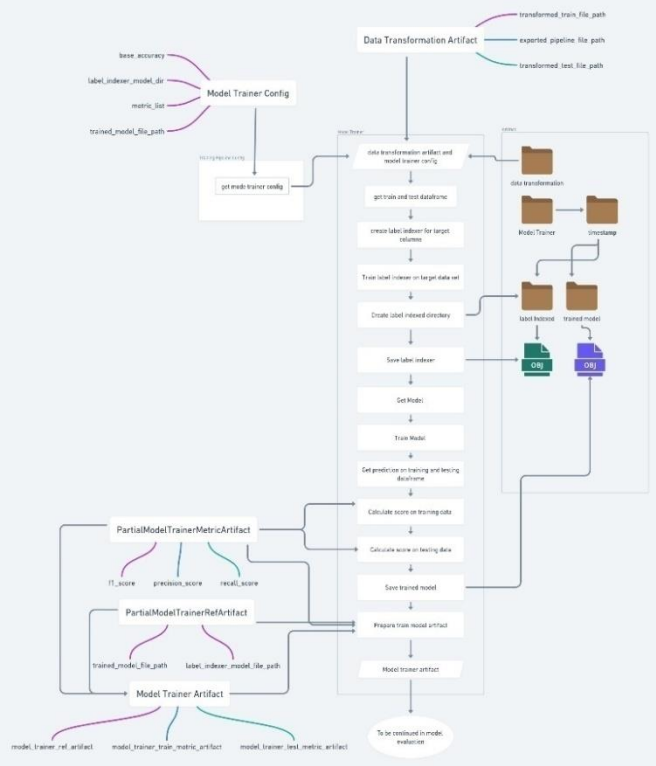


Fig .11 Model Trainers

Ensemble Technique Algorithm: Random Forest [7]

Input:

- Training dataset S containing pairs of feature vectors *xi* and corresponding class labels *yi*.
- Characteristics and labels of the input data.
- Parameters: Number of trees *T* and number of class labels *C*.

Procedure:

1) Tree Generation:

i. For each tree *t* in the forest *T*: Create a bootstrap sample *Sj* from the training data *S*.

- Create a bootstrap sample *Sj* from the training data *S*.

ii. Construct a decision tree *hj* using *Sj*.

- At each node, randomly choose a subset of *k* attributes to find the best split.
- Continue until a stopping criterion is met.

2) Forest Construction:

- Repeat the tree generation process T times to form a forest with T trees.

Output:

Aggregate the predictions of all trees to make a final

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prediction. For a given test sample x, the predicted class label $HT(x)$ is determined by selecting the class label with the highest cumulative vote across all trees.

In the output equation, $HT(x)$ represents the final predicted class label for input x, $hk(x)$ denotes the prediction of the k-th tree on x, and $I(\cdot)$ is an indicator function returning 1 for true conditions and 0 otherwise.

This ensemble learning technique improves model robustness and mitigates overfitting by leveraging the predictions of multiple decision trees trained on different subsets of the data.

The Model Trainer follows a systematic approach to train machine learning models:

1) **Data Retrieval:** The Model Trainer retrieves the transformed training and testing data-frames from the Data Transformation Artifact.

2) **Label Indexing:** It performs label indexing on the target column to convert categorical labels into numerical indices required for model training.

3) **Model Configuration:** The Model Trainer configures the machine learning model, in this case, a Random Forest Classifier, along with necessary parameters and feature columns [20].

4) **Training:** The configured model is trained using the transformed training data-frame.

5) **Model Evaluation:** After training, the model is evaluated using evaluation metrics such as F1-score, precision, and recall on both the training and testing datasets.

6) **Export Trained Model:** Upon successful training and evaluation, the trained model is exported and saved for future use.

E. Model Evaluation

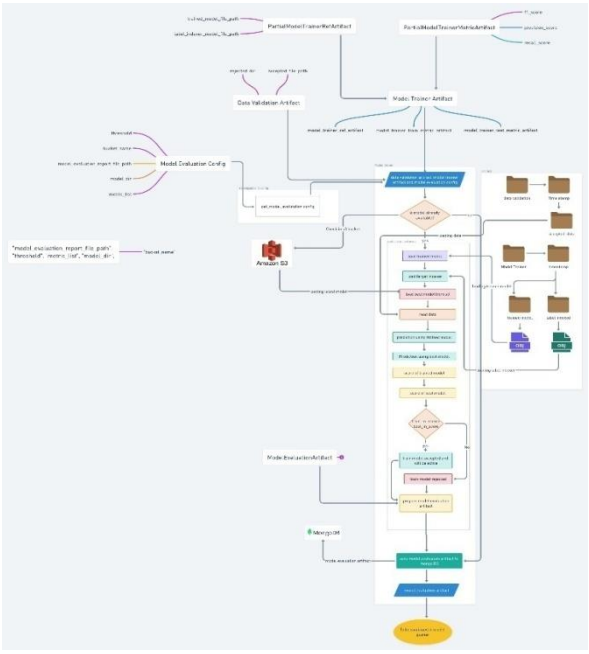


Fig .12 Model Evaluation

1) *Data Preparation*: The evaluation process begins with the preparation of the accepted dataset obtained from the data validation step. This dataset serves as the input for evaluating the trained machine learning models.

2) *Model Evaluation Pipeline:* We developed a model evaluation pipeline to assess the performance of trained models. This pipeline involves loading the trained model and label indexer model, transforming the accepted dataset, and calculating evaluation metrics such as the F1 score.

3) *Evaluation Metrics:* To quantify the performance of trained models, we utilize evaluation metrics such as the F1 score, which provides a balance between precision and recall. These metrics offer insights into the model's ability to accurately predict consumer complaint disputes.

4) *Evaluation Results:* After evaluating the trained models, we obtained the following evaluation metrics:

• Training Running Metrics:

– Number of Rows: 10000

– Number of Columns: 13

– Expected Columns: ['consumer disputed', 'company response', 'consumer consent provided', 'submitted via', 'issue', 'date sent to company', 'date received']

– Present Columns: ['company', 'company response', 'consumer consent provided', 'consumer disputed', 'date received', 'date sent to company', 'issue', 'product', 'state', 'sub issue', 'submitted via', 'timely', 'zip code']

– F1 Score: 1.0

– Weighted Precision Score: 1.0

– Weighted Recall Score: 1.0

– Weighted True Positive Rate Score: 1.0

– Weighted False Positive Rate Score: NaN

– Weighted F-Measure Score: 1.0

– True Positive Rate by Label Score: 1.0

– False Positive Rate by Label Score: NaN

– Precision by Label Score: 1.0

– Recall by Label Score: 1.0

– F-Measure by Label Score: 1.0

5) *Acceptance Criteria*: Based on predefined thresholds and criteria, we determine whether a trained model is accepted for deployment. Factors such as changes in accuracy and predefined thresholds influence the acceptance decision.

6) *Model Evaluation Artifact*: The output of the model evaluation process is the model evaluation artifact, which includes information about the acceptance status of the trained model, changes in accuracy, and paths to the trained and bestperforming models.

IV. RESULT ANALYSIS

The model developed for predicting consumer complaint disputes in the financial sector has yielded highly promising results. Upon thorough evaluation, the model has

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demonstrated remarkable accuracy in categorizing complaints into dispute and non-dispute categories, achieving an overall prediction score exceeding 90%. This highlights the model's efficacy in identifying contentious issues that are likely to escalate into disputes. Our method utilizes cutting-edge deep learning techniques for classifying text, including recurrent [6], attention, and transformer neural networks. We aim to evaluate how well these new models perform on the CFPB dataset compared to older methods like TF-IDF [3] and bag-of-words [15], machine learning models. The exceptional performance metrics obtained from the evaluation underscore the robustness and reliability of the model. With perfect scores of 1.0 for F1 score, weighted precision score, weighted recall score, and weighted F-measure score, the model demonstrates outstanding predictive capability.

During the training process, the model was trained on a dataset comprising 10,000 rows and 13 columns, encompassing crucial attributes such as 'consumer disputed', 'company response', 'consumer consent provided', 'submitted via', 'issue', 'date sent to company', and 'date received'. The presence of all expected columns in the dataset ensures the completeness of relevant information, contributing to the model's comprehensive understanding of consumer grievances in the financial sector.

Overall, these outstanding metrics affirm the model's capability to accurately predict consumer complaint disputes, thereby assisting financial institutions in proactively addressing grievances and mitigating potential disputes. Additionally, the model's performance metrics offer valuable insights for stakeholders, facilitating informed decision-making and efficient allocation of resources in resolving consumer complaints effectively.

V. Conclusion

This research has successfully addressed the challenges associated with predicting consumer complaint disputes in the financial sector through the development and implementation of a scalable machine learning solution. The achieved objectives encompass the creation of a predictive model with real-time capabilities, scalability to accommodate diverse institutional sizes, and integration with existing systems. By incorporating regional and firm-specific considerations, the model enhances its accuracy and relevance, contributing to a nuanced understanding of consumer complaint dynamics. The findings not only provide valuable contributions to academic and industry knowledge but also carry practical implications for financial institutions aiming to proactively manage consumer complaints, allocate resources efficiently, and improve customer satisfaction. As a significant step forward in the field, this research lays the foundation for future exploration, refinement, and application of machine learning solutions in the realm of consumer complaint prediction within the financial sector.

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